

Civic Register

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FEATURES

Frontend Engineering at Palantir: Redefining Real-Time Map Collaboration

Palantir · Palantir Blog

How we built lightweight, real-time map collaboration for teams operating at the edge.

Frontend engineering at Palantir goes far beyond building standard web apps. Our engineers design interfaces for mission-critical decision-making, build operational applications that translate insight to action, and create systems that handle massive datasets — thinking not just about what the user needs, but what they need when the network is unreliable, the stakes are high, and the margin for error is zero.

This series pulls back the curtain on what that work really looks like: the technical problems we solve, the impact we have, and the approaches we take. Whether you're just curious or exploring opportunities to join us, these posts offer an authentic look at life on our Frontend teams.

In this blog post, Leon, a Full Stack engineer based in NY, shares how building Gaia's Follow Along mode — his first frontend project at Palantir — shifted his perspective on frontend engineering, while diving into the unique challenges of enabling real-time collaboration at scale.

Before joining Palantir, I thought frontend engineering was mostly about adding pixels, pushing them around, and making things look nice. While that's certainly a component of frontend development, it's nowhere close to the complete picture of what frontend engineering looks like here. At Palantir, frontend engineers are deeply involved in solving complex, novel problems that demand creativity, technical rigor, and close collaboration across teams.

My perspective didn't change overnight — it changed through first-hand experience. Gaia Follow Along Mode was one of my first projects at Palantir, and it perfectly captured just how different frontend engineering is here.

Palantir Gotham is Palantir's platform for integrating, analyzing, and operationalizing complex data in mission-critical defense and intelligence environments. Used by American and allied forces worldwide, Gotham enables teams to collaborate seamlessly, share situational awareness, and make informed decisions — even in challenging operational contexts.

At the heart of Gotham's geospatial capabilities is Gaia, our collaborative map application. Gaia enables teams to analyze and share geospatial data in real time, providing a common operating picture across distributed environments. The Follow Along feature in Gaia allows users to "follow" another user's actions — tracking their cursor, zooming, panning, and selection activity as it happens.

This feature is particularly valuable in defense and other mission-critical environments, where teams are often distributed across remote bases and must operate under severe bandwidth constraints and high network latency.

Traditional screen sharing is often impractical or even impossible in these settings; even when video teleconferencing is available, the resulting screen shares are low resolution or laggy. Operational users need a lightweight, reliable way to share situational context — one that doesn't consume precious bandwidth, is resilient to latency, and doesn't require high-end connectivity.

Follow Along is a feature that addresses the shortcomings of traditional screen sharing by transmitting only essential state (such as map position, cursor location, and other select information), and sending updates only as changes occur, rather than continuously streaming every frame of the screen. This approach enables real-time collaboration without the heavy network load of true screen sharing.

On the surface, it might look like a simple feature, but building it required thoughtful engineering across multiple dimensions:

Real-time collaboration: Developing a real-time collaborative feature introduced several unique challenges.

Shared state: We needed to reliably synchronize user actions across multiple clients, ensuring that every interaction was reflected almost instantly for all participants. To achieve this, Follow Along leverages an internal shared-state streaming framework, along with libraries such as Redux, to track and update specific map and selection properties in real time. Each client subscribes to updates for the relevant artifact, and only users with the appropriate permissions can participate.

Ordered updates: To guarantee consistency, we introduced a monotonically increasing update index for cursor events, ensuring updates are processed in order and preventing jumpy or inconsistent cursor movement for followers.

Follower chaining and control: We had to handle follower chaining and prevent circular follows (where users could end up following each other in a loop), and ensure seamless hand-offs when a follower wanted to take control.

Disconnection handling: We had to gracefully handle disconnections. For example, if the leader disconnected, the Follow Along session would automatically end for all followers. However, to account for poor network connectivity and intermittent disconnections, we implemented a timeout that waits 15 seconds after the leader disconnects before ending the session for the follower.

Performance: Our users are often deployed in austere environments, where bandwidth is low, latency is high, and compute and hardware resources are limited. To maintain a smooth and responsive experience, we made careful optimizations to minimize resource usage and maximize reliability.

Selective data streaming: Given the restrictive network bandwidth, we had to carefully scrutinize the data included in our updates. While our maps can support large-scale

How robust is our accelerometer?

group · RealClimate

Guest commentary from Nathan Lenssen (Colorado School of Mines)

A new analysis of historical temperatures suggests that things are getting warmer faster, but what does it mean for the future?

As I'll discuss further below, there was no detectable acceleration in the raw GMT series – this doesn't mean there isn't any, but that the noise (internal variability etc.) doesn't allow us to see if there is clearly. Thus, the study has detected an acceleration in the rate of warming of inferred long-term trends – which we can pretty confidently attribute to anthropogenic effects. This study has understandably gotten substantial attention in the media. Here, I will outline what I think we have learned from this study, what this means for our understanding of the current state of the climate system, and what it means for projections of climate change (Hint: not much).

FR26 make three contributions in this recent work: (1) the production of an “adjusted” GMT series that removes statistically estimated impacts of a few short term changes in GMT, to hopefully leave just the warming associated with changes in anthropogenic forcings, (2) the detection of an acceleration in the rate of warming on this series using three different statistical methods, and (3) a forecast that 1.5°C warming will be reached by 2030. The methods used here are generally sound, particularly by engaging with the state of the art in changepoint detection methods as one of the methods for acceleration detection (Beaulieu et al. 2024). The figure below shows the three statistical methods for detecting changes in trend, all of which provide statistically significant evidence that the recent trend is faster than previous trends.

Figure 1: Figure 3 from FR26 showing the significant changes in rate (y-axis) as detected by three methods using the Berkeley Earth global mean temperature series.

Given the assumptions made by the authors, this provides statistically robust evidence that acceleration has been detected. On first glance, this may be surprising or alarming as, to the zeroth order from our understanding of the Earth's system's response to CO₂, we expect a roughly generally linear warming in GMT given the exponential rise in CO₂ due to the log-scaling of GMT with CO₂. Acceleration could be the result of the decrease of cooling anthropogenic forcings (as is hypothesized for some regional accelerations detected in Beaulieu et al. 2024) or substantial feedbacks/tipping points that are causing the Earth to warm faster than the simple CO₂ forcing physics dictates. Note though that the climate models that are used to inform our future projections also expect an acceleration around now (of course, given the assumptions that went into them).

However, as the authors point out, their method of ‘removing’ ENSO could be improved (for instance, Compo and Sardeshmukh (2010)), and there is still some imprint of natural climate variability in their adjusted time series. Note that an estimate of the “true” natural variability of the climate system, and correspondingly the “true” forced response, is one of the white whale problems in climate science! FR26 does an credible, but necessarily imperfect, job of isolating the forced response, but don't account for this uncertainty in their statistical tests.

Note though that the climate models that are used to inform our future projections also expect an acceleration around now (of course, given the assumptions that went into them).

group

While we can't know the true internal variability perfectly, we have climate models which provide an estimate of this variability. The figure below shows that the CMIP6 models (screened for a likely Transient Climate Response (TCR)) have a spread that fully contains the observed climate signal. Notably, the ensemble mean of these models demonstrates a slightly greater than linear warming (minus the effects 1991 eruption of Mount Pinatubo).

Figure 2. CMIP6 model SAT (with 95% spread) (historical + SSP245 after 2014), with a screening for likely TCR (1.4-2.2°C).

We can look at this more closely. If we look at the trends in individual model simulations for the last 13 years (2013-2025) and the 13 years before that (2000-2012), on average, the models show a slight acceleration over the same period highlighted by FR26 (0.18°C to 0.30°C). However, while there is a difference in the mean of these distributions, they are not clearly separate. This shows that, at least in model land, the acceleration in trend (given the internal variability and model uncertainty) is going to be difficult to detect. Note that comparisons between the models and the real world are complicated by any divergences in the forcings in the scenarios (designed more than 15 years ago) and what actually happened (Hunga Tonga, the IMO regulations, Chinese aerosol decreases etc.).

Figure 3. CMIP6 trends over the last 13 years (red) and the 13 years before (black) (using the screened simulations), along with the estimate trends from FR26 over (roughly) the same periods.

So where does this leave us? There is no detectable acceleration in the raw observed GMT, but there is an acceleration in GMT when removing the linear effects of ENSO, volcanoes, and solar variability, and there is slight acceleration in GMT when estimated using a multi-model ensemble of climate models. John Kennedy recently discussed some of these results in the context of FR26, expanding to a wider discussion of the implications for future projections. He highlights

NASA Selects Intuitive Machines to Deliver Artemis Science, Tech to Moon

Ivry Artis · NASA Breaking News

6 min read

Preparations for Next Moonwalk Simulations Underway (and Underwater)

A rendering of the Intuitive Machines larger cargo class lunar lander is pictured above with the Honeybee Robotics lunar rover (lower right) and the Australian Space Agency's Roo-Ver lunar rover (lower left).

NASA has awarded Intuitive Machines of Houston, \$180.4 million to deliver NASA-funded science and technology to the lunar surface as part of the agency's CLPS (Commercial Lunar Payload Services) initiative and Artemis program. This lunar delivery, which includes seven payloads — five of them NASA's — is expected to increase understanding of the chemical composition and structure of regolith, as well as the radiation environment in and around the South Pole region. This science will continue to build a sustainable human presence by future Artemis missions.

"NASA continues to progress lunar science and exploration by enabling commercial lunar landings," said Joel Kearns, deputy associate administrator for exploration, Science Mission Directorate, at NASA Headquarters in Washington. "These science and technology investigations aim to support long-term sustainability and contribute to a deeper understanding of the lunar surface, test technologies, and prepare for future human missions at the South Pole."

Intuitive Machines is responsible for delivering end-to-end payload services to the lunar surface, targeted to land at the Moon's South Pole region in 2030. This is the fifth CLPS contract for the company, which has delivered payloads to the Moon twice with their IM-1 and IM-2 missions.

"As NASA prepares to send humans and more robotic missions to the Moon, regular CLPS deliveries will provide a better understanding of the exploration environment, accelerating progress toward establishing a long-term human presence on the Moon, setting the stage for eventual human missions to Mars," said Adam Schlesinger, manager of the CLPS initiative at NASA's Johnson Space Center in Houston.

The rovers and instruments, totaling 165 pounds (75 kilograms) in collective mass include:

Stereo Cameras for Lunar Plume Surface Studies (SCALPSS) will use enhanced stereo imaging photogrammetry, active illumination, and ejecta impact detection sensors to capture the impact of the engine exhaust plume on lunar regolith as the lander descends on the Moon's surface. This payload flew on both Intuitive Machines' IM-1 and Firefly Aerospace's Blue Ghost Mission 1 and captured first of its kind imagery. The high-resolution stereo images will aid in creating models to predict lunar regolith erosion and ejecta characteristics, which is important as bigger, heavier spacecraft and hardware are delivered to the Moon near each

other. Lead organization: NASA's Langley Research Center in Hampton, Virginia

Near-Infrared Volatiles Spectrometer System (NIRVSS) will observe light emitted or reflected by the lunar soil to help identify its composition. NIRVSS is designed to detect minerals and various types of ices that may be present. NIRVSS will also take high resolution images of the lunar soil and composition variability, which could help inform how ices interact with the lunar soil. The instrument successfully powered on and collected data while in flight on Astrobot's Peregrine Mission One in 2024. NIRVSS aims to measure the surface temperature at fine scales, which may help determine where ice can exist or remain stable. Lead organization: NASA's Ames Research Center in California's Silicon Valley

Mass Spectrometer for Observing Lunar Operations (MSolo) will characterize the makeup of volatiles (things that easily evaporate) in the environment around the lander following touchdown. The mass spectrometer demonstrated its gas analysis capabilities in lunar conditions during Intuitive Machines' IM-2 mission in 2025. MSolo measures low molecular weight volatiles, which can be used as resources on the lunar surface. Lead organization: NASA's Kennedy Space Center in Florida

Lunar Vehicle Radiation Dosimeter system (LVRaD), a suite of four radiation detectors, is designed to quantify the radiation environment on the lunar surface and assess its potential impacts of radiation on biology and the human body in preparation for future human-related activities on the Moon. Additional sensors will investigate volatiles and geological resources that will help us plan for long-term exploration, as well as gain insights into the Moon's formation and solar system evolution. Lead organization: Korea Astronomy and Space Science Institute

Multifunctional Nanosensor Platform (MNP) is a highly compact and sensitive chemical analysis instrument designed to advance understanding of the lunar environment. It will investigate how exhaust plumes from a lander's engines interact with the lunar regolith by measuring volatile compounds over time and at varying distances from the landing site. These measurements will provide critical data to better understand plume-surface interactions and their effects, informing the design of safer, more sustainable landing systems and surface operations, directly supporting NASA's broader lunar exploration objectives. To enable these measurements, the MNP instrument will be integrated into the Australian Space Agency's rover ("Roo-ver"), a foundation services technology demonstration. The rover will showcase Australia's robotics capabilities, with the ability to traverse complex terrain and operate with limited human intervention. In doing so, Roo-ver will validate key mobility and autonomy technologies in the lunar environment while serving as the enabling platform for MNP's scientific objectives.

Lead organization for MNP: NASA's Goddard Space Flight Center in Greenbelt, Maryland

Gas price jump feels like a lot because it is

Nathan Yau · *FlowingData*

For NYT's the Upshot, Francesca Paris gives context to the recent spike in gas prices in the United States :

It's the second-largest four-week increase in at least 30 years — bigger than the one at the start of the war in Ukraine in 2022, or the ones associated with the post-recession surge of 2009 and the OPEC production cuts in 1999 .

The only bigger jump came after Hurricane Katrina in 2005, when gas supply fell significantly .

Rising everyday expenses probably don't help the situation. For those keeping track, 50 Cent is currently adjusted to 111 Cent as of this writing.

Tags: gas , prices , Upshot

BRIEFS

Scientific American, Scientific American: The past 11 years were the 11 hottest on record amid an increasing onslaught of climate-driven disasters, the World Meteorological Organization said in a new report

Trivago Tech, Trivago Tech: I'm happy to share that trivago has released a Prettier plugin which sorts import declarations in TypeScript and JavaScript modules for a given configured order. Throughout this article, I'll explain to you the motivation behind this Prettier plugin and how it works in detail.

Emily AddingtonKabo R. WaleEmily HorsburghMargot FargeasLeonidas SpathisWeronika LeśniakSaoirse FlavinPatricia T. Rimbi-David R. MarkSofia SandalliEster SerranoGavin BlackburnClément RegnaultPhillip D. WhitfieldJames P. R. ConnollyAndrew J. RoeNicky O'BoyleaSchool of Infection and Immunity, University of Glasgow, Glasgow G12 8TA, United KingdombDepartment of Biological Sciences, University of Botswana, Gaborone 0000, BotswanacDepartment of Microbiology, School of Genetics and Microbiology, Moyne Institute of Preventive Medicine, Trinity College Dublin, Dublin D02 A2H0, IrelanddMVLS Shared Research Facilities, University of Glasgow, Glasgow G12 8QQ, United KingdomeNewcastle University Biosciences Institute, Newcastle University, Newcastle-upon-Tyne NE2 4HH, United Kingdom, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceTo optimize within-host fitness, bacterial pathogens exploit intricate sensory mechanisms to precisely modulate gene expression in response to host-associated cues. D-serine, a host-produced metabolite enriched in urine inhibits the type 3 ...

Isaiah Colbert, Gizmodo: If you like 'Green Yuri,' check out Sal Jiang's new manga about a witch's foray into the magic of social media likes, thirst traps, and an enemies-to-lovers relationship with her internet troll.

Trivago Tech, Trivago Tech: At trivago we are building and using a Pattern Library which is based on Brad Frost's Pattern Lab adapted to our needs; our patterns are written in Twig. This Pattern Lab is based around Brad's Atomic Design, which is also something that we are embracing.

Keyu ChenYu DengTianming HuJichun LiXin-Yu HuangChengcheng ZhuHonggang ChenMeihua ChenLimin WuYonghui DengWenhe XieaDepartment of Chemistry, Shanghai Stomatological Hospital and School of Stomatology, State Key Laboratory of Coatings for Advanced Equipment, Shanghai Key Laboratory of Molecular Catalysis and Innovative Materials, Fudan University, Shanghai 200433, P. R. ChinabState Key Laboratory of Advanced Fiber Materials, Department of Materials Science and Engineering, College of Materials Science and Engineering, Donghua University, Shanghai 201620, P. R. ChinacInstitute of Energy and Materials Chemistry, Inner Mongolia University, Hohhot 010021, P. R. China, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceInspired by the periodic porous membrane and breathing process of the mammalian respiratory system, a steady nonequilibrium sensing process is established to achieve high-accuracy gas discrimination based on artificial periodic porous ...

Justin Carter, Gizmodo: The next stage of 'Steel Ball Run' will ride onto Netflix before 2026 ends. But how much will that be, and how will we get it?

Trey Alston, Consequence of Sound: Retired actress Kim Novak isn't feeling so euphoric about Sydney Sweeney portraying her in the upcoming film Scandalous! The star of Alfred Hitchcock's Vertigo recently spoke to The Times of London about the film that'll chart her relationship with Sammy Davis Jr. in 1957. In the interview, the two-time Golden...

Please click the link below to read the full article.

BBC Science & Environment, BBC Science & Environment: From a race with China to lunar discoveries, the US is investing time, effort and money to head to the Moon - and beyond.

Rubin DasguptaChristian SteinmetzgerAncy T. WilsonSatyaki ChatterjeeGunnar W. ReginssonSnorri Th. SigurdssonKatja PetzoldaDepartment of Medical Biochemistry and Microbiology, Science for Life Laboratory, Center of Excellence for the Chemical Mechanisms of Life, Uppsala University, Uppsala 751 23, SwedenbDepartment of Chemistry, Science Institute, University of Iceland, Reykjavik 107, Iceland, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. Significance-Understanding RNA structure and dynamics inside cells remains a key challenge in molecular biology. Here, we establish a site-directed spin labeling approach that enables targeted NMR signal enhancement of RNA in complex environments, using ...

Trivago Tech, Trivago Tech: When migrating your data to new technologies, validation of the data becomes challenging as your data structures might change. Rebase tries to make this easier while also giving your more flexibility on your data.

Scientific American, Scientific American: Sperm whales are known to socialize, but scientists were stunned when they saw a group of sperm whales gather as one of them gave birth

Langdon Hickman, Consequence of Sound: Black Label Society have released "Ozzy's Song" as part of the band's new album, Engines of Demolition, which is out today. The track was written penned by frontman Zakk Wylde in honor of his late friend Ozzy Osbourne. Wylde was a member of Ozzy's solo band for multiple stints, including...

Please click the link below to read the full article.

Scientific American, Scientific American: People are buying helmets, face masks, vests and beds that emit long-wavelength light. Beneath the hype, there is some interesting biology.

Yitzhak JaffeAndrew WomackKarine TachéMitya KletzermanNimrod MaromRon PalzurBo SunYichao ZhaoaSchool of Archaeology and Maritime Cultures, Department of Archaeology and the Zinman Institute of Archaeology, University of Haifa, Haifa 3103301, IsraelbDepartment of Anthropology, Furman University, Greenville, SC 29613cDepartment of Historical Sciences, Université Laval, Québec G1V 0A6, QC, CanadadAssociated Researcher, The Interdisciplinary Center for Archaeology and the Evolution of Human Behaviour (ICArEHB), University of Algarve, Faro, 8005-139, PortugaleShandong Institute of Cultural Relics and Archaeology, Shandong 250100, China, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceArchaeologists have long explored how states and political actors expand their territories, through conquest, colonization, trade. While peaceful

PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. Significance

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